

Literature Review: The Extension Boundary

From Coherent Charges to Probability Measures

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Abstract

A thematic survey of the literature bearing on the question: *when does observationally coherent data determine a genuine probability measure, and what additional commitments does this require?* The review is organized around four themes: (1) the boundary between finite additivity and σ -additivity, (2) Stone duality and the problem of realization, (3) value-definiteness and obstruction in non-distributive algebras, and (4) delay reconstruction as a source of observation algebras. Within each theme, classical results are placed alongside the philosophical, logical, and topos-theoretic perspectives that illuminate the boundary. Open problems and closed directions are documented where they arise. This is a working survey, not a journal submission; it occasionally marks my own contributions and dead ends alongside the literature.

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1 Introduction

The central question of this survey decomposes into three layers:

1. **Extension:** When do compatible finitely additive charges on a directed system of Boolean algebras determine a σ -additive probability measure?
2. **Realization:** When does a measure on a dual space (Stone space) descend to a measure on an underlying state space?
3. **Value-definiteness:** When does a probabilistic description admit simultaneous definite values for all observables?

Each layer requires a commitment the previous does not. The algebraic structure of the observation algebra constrains which commitments are available; distributivity is sufficient to make all three available.

The survey draws on measure theory, Boolean algebra, Stone duality, orthomodular lattice theory, topos theory, philosophy of probability, model theory, and dynamical systems. Coverage is not exhaustive within any one field; the aim is to document the results that bear on the three-layer decomposition, including the negative results that close off natural research directions.

Scope. The review covers:

- The classical extension theorems (Carathéodory, Kolmogorov, Choksi) and their algebraic characterization (Yosida–Hewitt, Kantorovich–Vulikh–Pinsker);
- The model-theoretic obstruction to first-order characterization of σ -additivity (Łoś, Hoover, Keisler);
- The philosophical context (de Finetti, Howson, Cox) and the logical coordinates (Frot, Espíndola, Caramello);
- Stone duality and the realization problem (Stone, Fremlin, Rao–Rao);
- Orthomodular lattice theory and contextuality (Kochen–Specker, Gleason, Pták–Pulmannová, McDonald–Bimbó);
- Delay embedding and reconstruction (Takens, Krieger, Sauer–Yorke–Casdagli);
- Sheaf-theoretic and topos-theoretic approaches to the extension problem (Zafiris, Biesel, Caramello, Vickers);
- Four open problems at the boundary.

2 The extension boundary: charges to measures

2.1 Classical extension theorems

The foundational results on extending finitely additive charges to σ -additive measures are well established:

- **Continuity from above.** A finitely additive charge ℓ on an algebra $\mathcal{A}$ is σ -additive if and only if $E_n \downarrow \emptyset$ implies $\ell(E_n) \rightarrow 0$ [25, §13, Theorem D].
- **Carathéodory extension.** A σ -additive premeasure on a semiring extends uniquely to a measure on the generated σ -algebra [25, §13], [5, Vol. 1, Ch. 1].
- **Kolmogorov extension.** Compatible finite-dimensional distributions on a product space extend to a unique probability measure under standard Borel/tightness conditions [4, Ch. 36].
- **Choksi inverse limits.** Compatible measures on an inverse system of compact Hausdorff spaces extend to the inverse limit [13, Theorem 1].

The **Yosida–Hewitt decomposition** [69] gives the structural anatomy: every bounded finitely additive charge decomposes uniquely as $\ell = \ell_c + \ell_p$, where ℓ_c is σ -additive and ℓ_p is purely finitely additive (PFA). The extension problem reduces to: when is $\ell_p = 0$?

2.2 The algebraic characterization

The deepest algebraic answer is Fremlin’s [21, §391] packaging of results going back to Kelley and to the theory of weakly (σ, ∞) -distributive algebras: a Boolean algebra $\mathfrak{A}$ admits a strictly positive σ -additive probability if and only if $\mathfrak{A}$ is

- (i) Dedekind σ -complete,
- (ii) weakly (σ, ∞) -distributive, and
- (iii) chargeable (admits a strictly positive finitely additive functional).

Kelley’s theorem [21, 391J] characterizes chargeability combinatorially.

Condition (ii) is the algebraic face of σ -additivity: it is a genuine lattice condition on how countable infima interact with arbitrary suprema, not a tautological restatement. This is the structural answer to the question of what property of the *algebra* (the site) — rather than the charges (the sections) — controls whether local data extend to a σ -additive global measure.

The boundary between chargeability and measurability was sharpened by Talagrand [61], who constructed a Maharam submeasure that is not equivalent to any measure, resolving Maharam’s 1947 control problem. This shows that the gap between conditions (iii) and (i)–(iii) together is genuine: an algebra can be chargeable without admitting any σ -additive probability.

2.3 Non-axiomatizability

A finitely additive charge on a Boolean algebra satisfies all first-order conditions that each individual measure in a family satisfies, by Łoś’s theorem [41]. Applied to Dirac masses $(\delta_n)_{n \in \mathbb{N}}$ on $\mathcal{P}(\mathbb{N})$ with a nonprincipal ultrafilter $\mathcal{U}$: each δ_n is σ -additive, but the ultraproduct charge $\ell(A) = \lim_{\mathcal{U}} \delta_n(A) = \mathbf{1}_{A \in \mathcal{U}}$ is the $\{0, 1\}$ -valued ultrafilter charge, which is purely finitely additive. Hence no first-order theory of Boolean algebras with charge can characterize σ -additivity (Hoover [28]; Fagin–Halpern–Megiddo [20]; see also Keisler [32]).

The same technique establishes non-axiomatizability at the level of pure Boolean algebra language (without charge predicates): the Lebesgue measure algebra and the Cohen algebra $\text{RO}(2^{<\omega})$ are both atomless, hence elementarily equivalent by Tarski’s classification, yet only the former admits a strictly positive σ -additive probability. (It satisfies all three KVP conditions; the Cohen algebra fails weak (σ, ∞) -distributivity.) This folklore observation appears implicit in Koppelberg [36, Ch. 18].

2.4 The philosophical context

De Finetti [15] grounds probability in *coherence* (the Italian *coerenza*, which translates equally well as “coherence” or “consistency”; the English literature settled on “coherence”). His cri-

terion — no finite combination of bets at the given rates guarantees a loss — characterizes exactly the finitely additive probability functions. De Finetti maintained that σ -additivity is a mathematical convenience with no operational justification: the Dutch Book argument for it requires extending the “finite sum of fair bets is fair” postulate to countable sums, which circularly entails σ -additivity [29, §§3–4].

Howson [29] makes de Finetti’s position precise by identifying two structural properties that de Finetti’s consistency shares with first-order logic:

- *Absoluteness*: consistency of P depends only on $P[E]$, not on which algebra E sits in.
- *Compactness*: inconsistency is detected by a finite subset of the assignments.

Both fail for σ -additivity. Howson identifies a missing “completeness theorem telling us that the rules of probability extend to finite but not countable additivity” [29, p. 17].

Cox’s axioms [14] produce finitely additive probability from three scale-free structural principles plus propositional logic. Howson notes [29, §6]: “Cox’s result does not extend to endorsing countable additivity. There are infinitary propositional languages... $L_{\omega_1, \omega}$... but there is no natural extension of Cox’s proof which would exploit that additional structure.” Ross [54] argues from the opposite direction that “all roads lead to violations of countable additivity,” reinforcing the view that σ -additivity requires commitment beyond coherence.

2.5 Logical coordinates

Three results locate the extension boundary within formal logic and topos theory.

Frot (2013). Deligne’s theorem (“coherent topoi have enough points”) is equivalent to Gödel’s completeness theorem for first-order logic [22]. The dictionary: models = points of the classifying topos; consistency = non-triviality; coherent theory = finitary axioms. Finite additivity is a coherent condition; σ -additivity is not (it requires countable universal quantification). So Deligne’s theorem guarantees the existence of finitely additive charges but is silent about σ -additive measures.

Espíndola (2020). Generalizes Deligne to κ -geometric logic [19]: at $\kappa = \omega_1$, σ -additivity becomes expressible. Property T (an exactness condition combining strong distributivity and dependent choice) is the structural content at each level. At $\kappa = \omega$, Property T is automatic, recovering Deligne. At $\kappa = \omega_1$, it is a genuine restriction — but applied to the site of finite Boolean subalgebras where σ -additivity lives, Property T reduces to σ -completeness of the underlying algebra plus uniqueness of the Carathéodory extension. Both are classical conditions already assumed in any CE setup. The machinery does not produce new content at this specific site.

Caramello (2018). The bridge technique [9] transfers properties between syntactic and semantic sites via Morita equivalences. But σ -additivity cannot be expressed as a geometric sequent (which permits only finite conjunction and small disjunction), so it cannot appear as a quotient of any geometric theory in Caramello’s framework.

2.6 Synthesis: five perspectives on the same boundary

A remarkable convergence emerges. Five independent traditions — operating with different vocabularies, motivations, and proof techniques — identify the same structural boundary:

1. **Philosophy of probability** (de Finetti, Howson): consistency (finite additivity) has absoluteness and compactness; σ -additivity has neither. Cox’s axioms produce finite additivity only.
2. **Logic and topos theory** (Frot, Espíndola, Caramello): Deligne (coherent topoi have enough points) = Gödel completeness = finitary completeness. σ -additivity escapes co-

herence. The ω_1 -coherent level can express it, but the machinery collapses to classical conditions on the relevant site.

3. **Algebraic measure theory** (Fremlin, KVP): weak (σ, ∞) -distributivity is the lattice condition that forces σ -additivity, answering Howson’s question in algebraic language.
4. **Model theory** (Łoś, Hoover, Keisler): ultraproducts kill first-order axiomatizability. The class of σ -additive structures is not elementary.
5. **Conglomerability and imprecise probability** (Seidenfeld et al. [56], Walley [66]): conglomerability — the requirement that conditional probabilities given a partition constrain the unconditional probability — is the operational face of σ -additivity. Schervish–Seidenfeld–Kadane show its failure is pervasive under finite additivity; Walley develops the imprecise-probability framework where the distinction is central.

No published work unifies these five perspectives under a single theorem. Whether such a unification would constitute genuine mathematics or terminological packaging is an open question. At minimum, the convergence confirms that the finite/countable boundary is a robust structural phenomenon, not an artifact of any particular framework.

2.7 Sheaf-theoretic approaches: closed directions

Several natural attempts to reformulate σ -additivity as a sheaf or descent condition have been investigated and found to yield dictionary translations rather than theorems:

- **Zafiris (2006)**. The Grothendieck topology J (epimorphic families) on the site of Boolean subalgebras does not grade covers by cardinality [70]. Finite, countable, and uncountable covers satisfy J uniformly; the finite/countable boundary is invisible.
- **Biesel (2024)**. Measures form a sheaf for finite covers [2, Theorem 8]; probability measures likewise [2, Theorem 11]. Countable covers are explicitly excluded [2, Remark 7]. The step from finite to countable covers is exactly the σ -additivity boundary, but this is tautological: finitely additive = sheaf for finite covers; σ -additive = sheaf for countable covers.
- **Caramello (2018)**. The bridge technique cannot express σ -additivity (§2.5).
- **Vickers (2008)**. The localic theory of valuations [65] provides the framework for formulating “a valuation is supported on the sublocale of principal ultrafilters” as a frame-internal condition, but this formulation has not been carried out. The earlier textbook [64] develops locales and sublocales without addressing measures.

The pattern is consistent: every finitary framework (sheaf-theoretic, topos-theoretic, locale-theoretic) either cannot see the finite/countable boundary or restates σ -additivity as a definition. The one framework that can express it (Espíndola’s ω_1 -geometric logic) collapses to classical conditions on the relevant site.

2.8 Open problems at the extension boundary

2.8.1 The σ -complete characterization pattern

Every known characterization theorem for when a Boolean algebra carries a strictly positive σ -additive probability presupposes Dedekind σ -completeness:

- Kelley [33], Theorem 9: complete + weakly distributive + chargeable.
- KVP [21, 391D]: Dedekind σ -complete + weakly (σ, ∞) -distributive + chargeable.

- Balcar–Jech–Pazák [21, 539N] (under p-ideal dichotomy): Dedekind complete + ccc $\Rightarrow$ Maharam algebra.
- Kalton–Roberts [21, 392F–G]: uniformly exhaustive submeasure + Dedekind σ -complete.
- All negative constructions (Argyros [1], Souslin line under $\neg$ SH, Farah–Veličković, Talagrand [61]) also operate in the σ -complete row.

The non- σ -complete regime has no analogous characterization — only piecewise existence results for specific construction families. This makes Strategy D (Problem 2.1 below) structurally novel: not “unstudied,” but *no measure-existence characterization reaches the non- σ -complete non-atomic regime*. Confirmed across ≥ 6 sources: Fremlin ch39, ch53, ch54; Kelley [33]; Argyros [1]; Džamonja–Plebanek [18].

2.8.2 Strategy D

Open Problem 2.1 (Measure-free Boolean algebra — Strategy D). Does there exist a Boolean algebra B satisfying:

- (i) B is non- σ -complete (equivalently, $\text{St}(B)$ is not basically disconnected);
- (ii) B is non-atomic (equivalently, $\text{St}(B)$ has no isolated points);
- (iii) B is measure-free: B admits no strictly positive σ -additive probability (equivalently, $\text{St}(B)$ carries no strictly positive Radon probability measure)?

Here σ -additivity on a non- σ -complete B is interpreted for disjoint countable families whose join exists in B .

On the clopen algebra of any Stone space, σ -additivity is vacuous by compactness; the real question concerns Radon measures on the Stone space itself.

Near-miss constructions. Table 1 summarizes the known candidates and which Strategy D conditions they satisfy.

Construction	(i) atomless	(ii) non- σ -c.	(iii) measure-free	Notes
Argyros [1]	✓	×	✓	under CH; βY route
Kunen L-space [45]	✓	✓	×	has s.p. measure by design
Souslin line (under $\neg$ SH)	✓	×	✓	consistency-only
Gaifman algebra	×	×	✓*	* no FA measure
Todorčević/D-P [18]	×	?	✓	ZFC; has atoms

Table 1: Near-miss constructions for Strategy D. Each fails at least one of the three required conditions.

Three structural patterns emerge:

1. The Stone–Čech compactification route (Argyros [1]) naturally produces basically disconnected spaces, i.e., σ -complete Boolean algebras. This is a systematic obstruction: the standard technique for building measure-free spaces also forces σ -completeness.
2. The L-space route (Kunen [45]) is consistency-only: under $\text{MA} + \neg\text{CH}$, no compact L-spaces exist. Even where they do exist, the construction’s purpose is to *produce* a strictly positive measure.
3. The $\mathcal{P}(\omega)/\text{fin}$ subalgebra route (Todorčević [62], Džamonja–Plebanek [18]) avoids σ -completeness but produces atoms (see below).

Every known ZFC measure-free Boolean algebra is either σ -complete or has atoms.

The Todorčević example. Džamonja–Plebanek [18, Theorem 3.1] extract from Todorčević’s Theorem 8.4 [62] a subalgebra $\mathfrak{A}$ of $\mathcal{P}(T)/\text{fin}$ generated by a family $\{T_a : a \in A\}$, where $A \subseteq Z$ (zero-density subsets of $\mathbb{N}$) is well-ordered under $\subseteq^*$ with order-type $\text{add}(\mathcal{N})$, and $T_a = \{(s, m) \in T : a \cap m \subseteq s\}$. They prove: $|\mathfrak{A}| = \text{add}(\mathcal{N})$, $\mathfrak{A}$ is ccc but not σ -centred,

and $\mathfrak{A}$ carries no strictly positive measure. The last follows from Lemma 3.2: the generating family satisfies $a \leq b$, $b \leq a$, or $a \cdot b = 0$ for $a, b \in \mathcal{G}$; combined with non- σ -centredness, this forces any strictly positive measure to assign positive mass to finitely many maximal chains, a contradiction.

However, the generators $\{[T_a] : a \in A\}$ form a chain in $\mathcal{P}(T)/\text{fin}$: for $a \subseteq^* b$, replace a by $a \cap b$ (same $\subseteq^*$ -class, since A is closed under finite changes within Z) to get literal $a' \subseteq b$, whence $T_b \subseteq T_{a'}$, so $[T_b] \leq [T_{a'}] = [T_a]$. A Boolean algebra generated by a single chain is an interval algebra. Since A has order-type $\text{add}(\mathcal{N})$ (an ordinal), immediate-successor pairs $a \prec a'$ produce atoms $[T_a] \setminus [T_{a'}]$ in $\mathfrak{A}$. Therefore $\mathfrak{A}$ fails condition (ii) — it is not atomless.

Consistency results. Džamonja–Plebanek [18, Corollary 2.8] prove: under $\text{MA} + \neg\text{CH}$, every atomless ccc Boolean algebra of size $< \mathfrak{c}$ carries a strictly positive nonatomic measure. Corollary 2.11 gives the converse: under $\text{MA} + \neg\text{CH}$, for atomless algebras of size $< \mathfrak{c}$, ccc is equivalent to the chain condition of Theorem 2.9. These results push Strategy D toward set-theoretic independence.

Current assessment. Plebanek’s survey [46] confirms the exact parameter regime (non- σ -complete, non-atomic, measure-free) remains open. Strategy D is likely ZFC-independent, but this is not proved. No ZFC construction or ZFC obstruction is known.¹

2.8.3 Ultralimit representability

Open Problem 2.2 (Ultralimit representability — reduces to Problem 2.1). Let B be a Boolean algebra, $\mathcal{U}$ a nonprincipal ultrafilter on an index set I , and $(\mu_i)_{i \in I}$ a family of σ -additive probabilities on B . Call a purely finitely additive charge ℓ on B *ultralimit-representable* if $\ell(b) = \lim_{\mathcal{U}} \mu_i(b)$ for all $b \in B$.

For which Boolean algebras B is every purely finitely additive charge ultralimit-representable?

Settled in all regimes except Problem 2.1:

- *Atomic* B (e.g. $\mathcal{P}(\mathbb{N})$): every PFA charge is representable (ultrafilter charges via Diracs; the rest by Krein–Milman).
- *σ -complete* B : Nikodým convergence [17, IV.9.8] forces ultralimits to stay σ -additive. No PFA charge arises.
- *Non- σ -complete, non-atomic* B : failure of representability is witnessed by a measure-free direct-product factor, reducing to Problem 2.1.

2.9 The operational boundary: estimation theory meets the finite/countable gap

The finite/countable boundary is epistemically inaccessible from finite data: no finitary test — first-order sentence, finite cover, sheaf condition on finite refinements — separates σ -additive measures from purely finitely additive charges (§2.6). Model-theoretically, this is Lindström’s theorem: first-order logic is the strongest logic with compactness, and compactness is what ultraproducts exploit to produce PFA charges from σ -additive ones.

Yet the boundary has operational consequences that are detectable from finite samples. The connection runs through minimax estimation under contamination.

The Yosida–Hewitt / contamination correspondence. Huber’s ε -contamination model assumes observations from $(1 - \varepsilon)f + \varepsilon g$, where f is the target density and g is an unknown

¹**Superseded 2026-06-18.** This regime in fact has a *positive* ZFC answer and was never field-open: Gaifman 1964 (PJM 14(1):61–73, Thm 2.2 + property (†)) already exhibits an atomless Boolean algebra with no strictly-positive *finitely*-additive measure, in ZFC. The non- σ -complete + atomless conditions are trivial; the literature records these algebras only in their *completed* (extremally/basically disconnected) form, which is why the regime “looked” open. “Strategy D” was this programme’s private name; retired.

contamination. This is structurally parallel to the Yosida–Hewitt decomposition $\mu = \mu_c + \mu_p$: the σ -additive part μ_c is the “target” and the PFA part μ_p is the “contamination” — a component that passes all finite consistency checks but does not concentrate on any σ -additive structure.

Liu–Gao [40] establish the minimax rate for density estimation under contamination (Theorem 2.1):

$$\mathcal{R}(\varepsilon, \beta_0, \beta_1) \asymp [n^{-2\beta_0/(2\beta_0+1)}] \vee [\varepsilon^2(1 \wedge m)^2] \vee [n^{-2\beta_1/(2\beta_1+1)} \varepsilon^{2/(2\beta_1+1)}],$$

where β_0 is the target smoothness, β_1 the contamination smoothness, ε the contamination proportion, and m the local contamination level. Three terms:

1. $n^{-2\beta_0/(2\beta_0+1)}$: the classical minimax rate (no contamination);
2. $\varepsilon^2(1 \wedge m)^2$: an n -independent floor determined by the contamination mass and level;
3. $n^{-2\beta_1/(2\beta_1+1)} \varepsilon^{2/(2\beta_1+1)}$: an interaction term coupling sample size to contamination structure.

Term (2) is the key: an irreducible estimation floor that no amount of data can overcome, proportional to the contamination mass. In the Yosida–Hewitt reading, this is the cost of the PFA component: mass that is invisible to every finite refinement but creates a permanent bias in any estimator.

Plateau detection. The transition between the n -dependent regime (term 1) and the ε -dependent floor (term 2) is detectable from the data. Lepski’s method [39] and its descendants (Goldenshluger–Lepski) perform adaptive bandwidth selection that halts when increasing resolution stops improving the estimate — i.e., when the estimator hits a plateau. The cause of the plateau (contamination mass, resolution limit, or model misspecification) is not identifiable from finite data; but the plateau itself is.

This is the precise operational sense in which the finite/countable boundary is “felt” without being “seen”: the convergence rate of an estimator encodes which side of the boundary governs the dominant error term. If σ -additive, the rate improves with n ; if the PFA component dominates, the rate plateaus.

Intrinsic dimension. In the classical rate $n^{-2s/(2s+d)}$, the exponent d is traditionally the ambient dimension. The intrinsic-dimension literature replaces d with an effective dimension adapted to the data geometry: Kpotufe [37] shows that k -NN regression achieves $n^{-2s/(2s+\rho)}$ with ρ the local intrinsic dimension; Yang–Barron [68] characterize minimax rates via metric entropy, which encodes effective dimension. The programme’s D_Λ (observational resolution dimension, the ratio of distinguishability growth to valuation growth) is the same substitution in the language of valued refinements. This is not new mathematics; it is a translation confirming that the rate-theoretic content of D_Λ is the standard metric-entropy exponent.

The recursive undermining. One might ask whether a higher-order information-theoretic criterion could *detect* the boundary (not just feel its consequences). The answer is no, by essentially the same compactness argument that operates at first order. Any criterion weaker than σ -additivity is satisfied by PFA charges (they pass all finite tests); any criterion strong enough to exclude PFA charges quantifies over all countable sequences and is equivalent to σ -additivity. There is no intermediate logic between first-order (compactness, can’t see the boundary) and full second-order (no compactness, sees it by brute force) that both has compactness and can express σ -additivity (a consequence of Lindström’s theorem applied to probability structures).

The Kolmogorov extension mechanism. One might hope that the structural constraint of finite additivity—the PFA component must marginalize consistently across resolutions—would give a multi-resolution estimator leverage unavailable against arbitrary contamination at *fixed* resolution. It does not. At any fixed finite collection of resolution levels $G_1, \dots, G_K$, the restrictions of a PFA charge form a consistent projective family indistinguishable from the marginals of some σ -additive measure (by Kolmogorov extension on finite systems). The

contamination floor ε^2 is therefore the same whether the contamination is genuinely PFA or arbitrary, at any single resolution snapshot.

However, this indistinguishability is *pointwise*, not *asymptotic*. A PFA charge on $B = \bigcup_k G_k$ cannot have σ -additive marginals at all levels simultaneously—if it did, it would agree with the σ -additive extension on all of B , contradicting pure finite additivity. In the estimation setting where $k^*(n) \rightarrow \infty$ as $n \rightarrow \infty$, the PFA component’s singularity (Yosida–Hewitt) eventually forces its marginals outside any fixed smoothness class, making the contamination fraction ε consistently estimable in the limit [44, 11]. The boundary is invisible at each finite stage but visible in the *trajectory* of increasing resolution—the precise sense in which it is “felt” operationally.

Synthesis. The estimation-theoretic perspective does not resolve the finite/countable boundary; it translates the boundary into a rate-theoretic statement. The translation connects three literatures that do not normally cite each other:

- *Probability foundations* (de Finetti, Howson, Seidenfeld): σ -additivity is not forced by coherence; the boundary is philosophically open.
- *Robust estimation* (Huber, Liu–Gao, Lepski): contamination creates an n -independent floor; adaptive methods detect the plateau.
- *Model theory* (Lindström, Łoś, Hoover): compactness prevents first-order detection; the boundary is logically inaccessible.

The convergence is that all three traditions agree on the same structure: the boundary cannot be detected from finite data, but the *rate* at which finite data converges encodes which side of the boundary dominates. Under conditions where the PFA component is small relative to the observation resolution — the operational content of countable additivity in the programme’s framework — the distinction between “is σ -additive” and “looks σ -additive at every accessible scale” is vacuous.

3 Stone duality and realization

3.1 Classical framework

Stone representation. Every Boolean algebra B is isomorphic to the clopen algebra of a compact, Hausdorff, totally disconnected space $\text{St}(B)$ (its Stone space) [31].

Charge–measure correspondence. Every finitely additive charge on B corresponds to a regular Borel (Baire) measure on $\text{St}(B)$ [21, §326], [25, §53].

Support characterization. A charge on B is σ -additive if and only if the corresponding Stone-space measure vanishes on meagre Baire sets, equivalently assigns full mass to countably complete ultrafilters [50, Theorem 10.5.3].

Principal ultrafilters. Every principal ultrafilter is countably complete. The converse holds for $\mathcal{P}(\kappa)$ when κ is below the first measurable cardinal (Ulam’s theorem). For non- σ -complete algebras (typical cylinder algebras), countably-complete non-principal ultrafilters can exist without large cardinals.

3.2 The realization problem

The Stone construction gives an *unconditional* object: a Baire probability measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$ always exists from compatible charges, by Stone representation combined with Choksi’s inverse limit theorem.

Descent to a state space Ω requires that $\hat{\mu}$ be concentrated on principal ultrafilters: $\hat{\mu}(\text{pure}(\Omega)) = 1$, where $\text{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$. Under discriminability and the assumption that countably-complete ultrafilters are principal (which holds for σ -complete algebras below the

first measurable cardinal), concentration on principal ultrafilters is equivalent to σ -additivity of the component charges.

The three-level nesting captures the structure:

$$\text{pure}(\Omega) \subseteq \{\text{countably complete ultrafilters}\} \subseteq \text{St}(\mathcal{C}).$$

σ -additivity gives mass on the middle set (intrinsic to the algebra). Descent to Ω requires mass on the inner set (depends on the choice of realization).

3.3 The philosophical dimension

Which ultrafilters count as “realized” is additional structure over the algebra. In the Boolean case, the distinction between the Stone measure (unconditional) and its descent to Ω parallels van Fraassen’s [63] distinction between empirical adequacy and realist commitment: the Stone measure is the “empirically adequate” object; descent requires choosing which points are real. The identification of Stone-space construction with empirical adequacy (EA), descent with probabilistic realism (PR), and concentration on dispersion-free states with value-definite realism (VDR) has no precedent in the literature. It is a philosophical contribution connecting van Fraassen to quantum logic via McDonald–Bimbó duality, not a mathematical theorem.

4 Value-definiteness and quantum logic

4.1 Distributivity and ultrafilters

Boolean algebras admit 2-valued homomorphisms (ultrafilters) by Zorn’s lemma. These are the dispersion-free states: they assign definite values (0 or 1) to every proposition simultaneously.

The Kochen–Specker theorem [35] shows that for the lattice $L(H)$ of closed subspaces of a Hilbert space H with $\dim H \geq 3$, no dispersion-free state exists. Non-distributivity is the structural cause: $L(H)$ is orthomodular but not distributive, and the failure of distributivity blocks the lattice-theoretic argument that guarantees ultrafilters in the Boolean case.

Gleason’s theorem [23] completes the picture for $L(H)$, $\dim H \geq 3$: every σ -additive state is of the form $s(P) = \text{tr}(\rho P)$ for a unique density operator ρ . (The theorem fails for $\dim H = 2$, where dispersion-free states on $L(\mathbb{C}^2)$ do exist.) Value-definiteness is blocked in dimension ≥ 3 , but probabilistic realism (states as density operators) is available.

The Bub–Clifton theorem [7], [26] identifies the maximal Boolean subalgebra $D(\rho, R) \subseteq L(H)$ on which a given state is a mixture of dispersion-free states. This gives the precise boundary of value-definiteness within a non-Boolean algebra.

4.2 Orthomodular lattice duality

McDonald–Bimbó [42] establish a topological duality for orthomodular lattices analogous to Stone duality for Boolean algebras. A key structural difference: principal filters $P(A)$ are part of the dual structure in the OML case (unlike Stone duality where they are invisible). This means realization is *constrained* structure in the OML case, not *additional* structure as in the Boolean case. As of 2026, this duality has received essentially no attention in quantum foundations.

Gunji–Haruna–Uragami [24] show that gluing Boolean algebras via categorical pushout produces orthomodular lattices, generically non-distributive.

4.3 Topos approaches

Three programmes reformulate quantum logic in topos-theoretic terms:

- **Isham–Butterfield [30]:** the topos $\mathbf{Sets}^{\mathcal{V}(\mathcal{H})^{op}}$ of presheaves on the poset of commutative subalgebras. The spectral presheaf $\underline{\Sigma}$ assigns to each Boolean context its Gelfand spectrum; it has no global elements = Kochen–Specker.
- **Döring–Isham [16]:** Daseinisation maps a projection to a sub-object of $\underline{\Sigma}$ by approximating within each Boolean context. This goes *up* (approximating non-Boolean by Boolean); descent goes *down* (Stone space to realized points).
- **Heunen–Landsman–Spitters [27]:** Bohrfication encodes the same structure via spectral presheaves. More powerful than the single-dual-space approach (presheaf over contexts, not single Stone space) but connects less directly to van Fraassen’s framework.

Rédei–Summers [51] review quantum probability in the von Neumann algebra framework: classical probability = abelian von Neumann algebra; quantum = nonabelian; normal states = σ -additive.

4.4 The OML extension problem

4.4.1 State space universality

Shultz [57] proves that orthomodularity imposes *no* geometric constraint on state spaces: every compact convex subset of a locally convex space arises as the state space of some σ -orthomodular lattice. The construction uses Greechie pasting (G -spaces): points and frames with loop constraints. However, σ -additive state spaces can be geometrically degenerate — the Lebesgue measure algebra has a σ -state space with no extreme points, hence not affinely equivalent to any compact convex set [57, p. 327].

4.4.2 Exotic logics and the Greechie pasting technique

Pták [47] constructs *exotic logics*: σ -orthocomplete OMPs admitting finitely additive states but no σ -additive states. The construction uses Greechie pasting: atom identifications across Boolean blocks create contradictory partition-of-unity constraints at the σ -additive level. Each block individually admits σ -additive states; the global pasting forbids them.

Navara [43] shows that regularity (P -regularity, inner regularity with respect to finite-dimensional projections) does *not* force σ -additivity on general OMPs. He modifies Pták’s construction to produce exotic logics on which all states are P -regular yet none are σ -additive. (The Beaver–Cook result that regularity forces σ -additivity is special to von Neumann algebras.)

Synthesis. Greechie pasting is the universal construction tool for non-Boolean phenomena: Pták [47] uses it to produce exotic logics, Shultz [57] to prove state-space universality, Navara [43] to kill regularity as an admissibility condition, and Wilce [67] surveys the technique systematically. The obstruction to σ -additive extension in the non-Boolean case is combinatorial/geometric (from the pasting), structurally different from the Boolean case where the obstruction is analytic (weak (σ, ∞) -distributivity).

4.4.3 The structural obstruction

Pták–Pulmannová [48] prove: if an orthomodular lattice possesses a unital set of subadditive probability measures, it must be Boolean (distributive). Conditions rich enough to force σ -additivity destroy the non-Boolean structure.

Related results exist under strong completeness conditions: Bunce–Wright [8] solve the Mackey–Gleason problem for von Neumann algebras (every bounded finitely additive measure on the projection lattice extends to a bounded linear functional), and Chetcuti–Dvurečenskij [12] use finitely additive states to characterize completeness of inner product spaces. Neither directly

addresses the FA-to- σ -additive extension question, and no result bridges the gap between “too weak” (allowing non- σ -additive states) and “too strong” (forcing Booleanity).

Open Problem 4.1 (OML extension). Let A be an orthomodular lattice and $s: A \rightarrow [0, 1]$ a state. Let $S_0(A)$ be the McDonald–Bimbó dual space with clopen $\perp$ -stable sets $\text{CO}(S_0(A))^\dagger$, and define $\mu(h(a)) = s(a)$ where $h(a) = \{x \in F(A) : a \in x\}$.

- (a) Under what conditions on A does μ extend to a σ -additive measure on the Baire σ -algebra of $S_0(A)$?
- (b) Does any condition strictly weaker than Booleanity suffice?

For $A = L(H)$ with $\dim H \geq 3$, Gleason gives the extension. For general OMLs, Gleason-type results are unavailable and Carathéodory does not apply (the clopens form an OML, not a Boolean algebra).

5 Delay reconstruction

5.1 Embedding theorems

Takens (1981). For a smooth compact manifold M with C^2 diffeomorphism T and generic C^2 observable h , the delay map $\Phi_L(x) = (h(x), h(Tx), \dots, h(T^{L-1}x))$ is an injective immersion for $L \geq 2 \dim M + 1$ [60].

Sauer–Yorke–Casdagli (1991). Prevalence-based generalization: the set of observables for which Φ_L is an embedding has full prevalence measure for $L > 2d_B(X)$ where d_B is the box-counting dimension [55].

Botvinick–Greenhouse et al. (2025). Measure-theoretic Takens theorem via optimal transport: if the pushforward measures converge in Wasserstein distance, the delay map is a measure-theoretic embedding [6].

5.2 Generating partitions and algebraic reconstruction

The ergodic-theoretic parallel: an observable σ -algebra equals the full σ -algebra mod μ if and only if the delay algebra is dense in L^2 , if and only if the delay map is a measure-theoretic embedding. This is Krieger’s generating partition theorem [38] in delay-map language (cf. Sinai [58] for the entropy definition, Rokhlin [53] for the metric).

The Rokhlin distance $\delta(\mathcal{O})$ measures how close the observable σ -algebra is to generating everything. Collision probability (=Rényi-2 entropy of the fibre decomposition) measures the resolution structure.

Geometric $\neq$ algebraic reconstruction. Collision convergence (geometric) and $\delta \rightarrow 0$ (algebraic) are genuinely inequivalent. A skew-product counterexample ($X = A^{\mathbb{Z}} \times B^{\mathbb{Z}}$, $T = \sigma \times \sigma$, $h(a, b) = a_0$) gives collision $= 2^{-(L+1)} \rightarrow 0$ (fibres shrink in the a -coordinate) but $\delta_L = 1/4$ for all L ($\{b_0 = 0\}$ is maximally unresolved). This is the one observation from this programme that I have not found in the prior literature, though it may well be known to ergodic theorists.

5.3 Practical diagnostics

- **False nearest neighbours (FNN):** embedding dimension selection [34], sensitive to noise [52].
- **Ragwitz–Kantz:** local prediction error minimization for (d, L) selection [49].
- **Billings–Voon:** residual autocorrelation as model validity test [3].
- **Casdagli:** deterministic vs. stochastic modelling discrimination [10].

Every practical criterion implicitly assumes a modelling commitment: deterministic criteria fail under noise because noise violates the commitment. Finite-sample convergence rates are $n^{-s/(2s+d)}$ [59]; elbow stopping via Lepski [39].

5.4 The reconstruction problem in programme context

The delay setting is the primary source of observation algebras for this programme: a dynamical system with a scalar observable produces a directed system of Boolean algebras (delay algebras at increasing embedding dimension). The three-layer question then becomes: under what conditions on the dynamics and observable do the resulting charges extend (σ -additivity), descend (realization), and determine definite values (value-definiteness)?

In the Boolean case (classical dynamics), all three are available in principle; the obstruction is purely at the extension level (σ -additivity). In the quantum case (non-commutative dynamics), value-definiteness is blocked by Kochen–Specker and the extension problem takes a different form (Problem 4.1).

6 Ergodic theory of finitely additive charges

The Yosida–Hewitt decomposition commutes with shift-invariance: if ℓ is a T -invariant charge, both ℓ_c and ℓ_p are T -invariant. The structure of the σ -additive part is the subject of classical ergodic theory (Birkhoff, von Neumann). The purely finitely additive part is far less understood.

Open Problem 6.1 (Extreme PFA shift-invariant charges). Let $T: \Omega \rightarrow \Omega$ be a measurable transformation on a standard Borel space. What are the extreme points of the convex set of T -invariant purely finitely additive charges on Ω ?

This connects to the structure of Banach limits and invariant means. The problem appears out of reach with current techniques.

7 Summary of the boundary

The extension boundary takes different forms in different algebraic settings:

Setting	Obstruction	Characterization	Status
Boolean, σ -complete	analytic	KVP (391D)	classical
Boolean, non- σ -complete	analytic/topological	Problem 2.1	open
OML, $L(H)$, $\dim \geq 3$	algebraic	Gleason	classical
OML, general	combinatorial	Problem 4.1	open

The Boolean and non-Boolean extension problems are structurally independent: the Boolean obstruction is analytic (weak (σ, ∞) -distributivity, or existence of measure-free factors), while the OML obstruction is combinatorial (Greechie pasting creating contradictory normalization). Neither subsumes the other. In the Boolean non- σ -complete case, every known ZFC measure-free construction is either σ -complete or has atoms (Table 1); the exact Strategy D parameter regime is likely ZFC-independent.²

Sheaf-theoretic, topos-theoretic, and locale-theoretic approaches to the Boolean extension problem converge on dictionary translations rather than theorems: the finite/countable boundary is invisible to finitary frameworks, and infinitary frameworks restate classical conditions. The OML extension problem has received less attention from these perspectives; the McDonald–Bimbó duality provides the natural framework but the extension question within it remains unasked in the literature.

²**Superseded 2026-06-18** — see the footnote at the “Strategy D is likely ZFC-independent” claim earlier: the regime has a positive ZFC answer (Gaifman 1964) and the Table’s “ σ -complete” entries read σ -completeness off the *completed*/Gleason-cover form, not the base algebra. Strategy D retired as a research target.

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